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Effect of flow and humidity on indoor deposition of particulate matter☆

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ABSTRACT

The removal of particulate matter (PM) is an important issue in public health and the global atmospheric environment. Various PM removal methods have been suggested to effectively remove PM particles. However, the effects of various factors on PM deposition are not completely clear. We quantitatively investigated the effects of flow and humidity difference in a closed chamber on PM deposition. To elucidate the parameters affecting the deposition of PM particles, PM removal efficiency and deposition constant were examined at different flow rates, flow directions, and relative humidity (RH) inside the closed system. The highest PM deposition rate was achieved under humid condition with the upward direction flow at a fan speed of RPM = 150. Mean velocity fields inside the test chamber were obtained by a particle image velocimetry (PIV) technique to quantitatively examine the effect of flow conditions on the PM deposition. The flow structure and RH inside the closed chamber have significant influence on PM deposition.

1. Introduction

The suspended particulate matter (PM) in the air is a critical hazard to health and to the global atmosphere (Dominici et al., 2014). PM_{10} and $PM_{2.5}$ are classified according to their diameters under 10 and 2.5 μm, respectively. In particular, they have adverse effects on pulmonary health, especially in individuals with preexisting lung diseases based on results from epidemiological analysis (Borm et al., 2006). Besides, cardiovascular diseases might be caused by the exposure to PM particles suspended in the air for a long time (Brook et al., 2010).

The importance of PM as a determinant of indoor air condition is strongly associated with particle concentration, chemical composition, and size distribution. Thus, many factors including surface roughness, flow pattern, and electrostatic force may contribute to indoor PM deposition (Lai, 2002; Mohan, 2016). Some researchers have reported that surface roughness affects the deposition rate of particles. Abadie et al. (2001) studied the effect of real wall textures on indoor PM deposition. A rough surface led to higher particle deposition than smooth surface. The effects of furniture and flow on PM deposition rate were also investigated (Thatcher et al., 2002). Experiments were conducted using three different indoor furnishing levels and four air flow conditions in an isolated room. The increase in the surface area and air speed enhanced the deposition rate of all particle sizes. In addition, the entry of outdoor PM into indoor was investigated (Rim et al., 2013). The indoor air in residential houses was investigated to quantify the particle deposition rate of particles in the size range from 0.015 to 6 mm (He et al., 2005). The deposition of particles onto indoor surfaces was numerically predicted with a modified Lagrangian method. The Lagrangian method was found to predict the deposition of indoor particles with a reasonable accuracy provided the near-wall turbulence and its interactions with particles (Zhang and Chen, 2009). To remove indoor PM, portable air-purifying devices were developed, and their PM particle removal performance was validated using the mass balance model (Lee et al., 2015; Sultan et al., 2011).

The other mechanism of PM deposition is associated with the hygroscopic characteristic of particles in response to relative humidity (RH). It has been reported that particle agglomeration by augmented RH enhances particle deposition based on its hygroscopic properties (Mikhailov et al., 2006; Montgomery et al., 2015). Recently, Ryu et al. (2019) verified a relationship between the removal of PM and RH using real plant and paper models.

In this study, the effects of internal flow and RH conditions in a

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closed chamber on PM deposition were experimentally investigated. To quantitatively analyze the main parameters affecting PM deposition, the PM removal efficiency and deposition constant were systematically examined at varying flow rates, flow directions, and RHs inside a test chamber. Particle image velocimetry (PIV) measurement was conducted to quantitatively verify the effect of flow condition inside the chamber on the PM removal performance. Results would provide basic information required for the design of an effective indoor air-purifying device.

2. Experimental apparatus and methods

2.1. PM measurement

The PM deposition experiments were conducted in two same test chambers of 2.7×10^{-2} m^3. Test chambers are made of stainless steel and glass to reduce electrostatic effect on PM deposition on the chamber surfaces (Fig. 1a). Burning incense was used as test particles. Smoke from burning incense consists mainly of gaseous oxide products and volatile organic compounds (VOCs) (Lin et al., 2008).

The distribution of PM concentrations measured inside the chamber were smaller than 5% and 1% for PM_{10} and $PM_{2.5}$, respectively, when the incense particles were injected and dispersed inside the chamber. The initial PM concentration was set as the total suspended particulate (TSP) index range of 1300 – 1600 μg/m^3. PM concentration was monitored for 2 h using a particle counter (TES-5200). Each monitoring interval was 1 min long with a flow rate of 2.84 L/min. Variations of RH and temperature in the test chamber were monitored simultaneously. After each experiment, the test chamber was cleaned with a dust-free paper towel, deionized water, and filtered air to maintain a particle-free environment for repetitive experiments.

Four repetitive experiments were conducted for each experimental condition. Among them, two experiments were conducted in each of two chambers. As summarized in Table S1, for dry chamber w/o flow case, the differences in PM concentrations for two chambers are 1.5% for PM_{10} and 3.1% for $PM_{2.5}$. They are less than the overall standard deviation of 1.7% for PM_{10} and 3.7% for $PM_{2.5}$. Therefore, it was confirmed that the measurement deviation between two test chambers is smaller than the overall standard deviation.

A CPU fan of 120 mm × 120 mm size was used to induce internal flow. The fan was fixed by a 3D printed structural frame inside the test chamber (Fig. 1a, right). Fan rotational speed changed from 0 RPM to 150 RPM with steps of 50 RPM. Even under no flow conditions, the structure was located inside the test chamber. The upward and downward flow conditions were established by reversing the fan direction at the same location in the test chamber. RH in the test chamber was regulated by adjusting the amount of water spray. RH condition was maintained for 2 h.

2.2. PIV measurement

PIV experiment for quantitative flow visualization was conducted in the same test chamber used for PM measurement. The schematic diagram of the PIV system utilized for measuring internal flow velocity fields inside the test chamber as shown in Fig. 1b. The PIV system consists of Nd:YAG laser, a delay generator, a CCD camera, optics for generating a laser sheet, and the data acquisition system (Kim et al., 2017). The same as the PM measurement test, incense smoke particles were employed as tracer particles for the PIV experiment. The mean velocity contours were obtained by averaging 900 pairs of instantaneous velocity fields under each experimental condition. Using the PIVview (PIVview, PIVTEC, Germany), PIV analysis was performed. A cross-correlation PIV

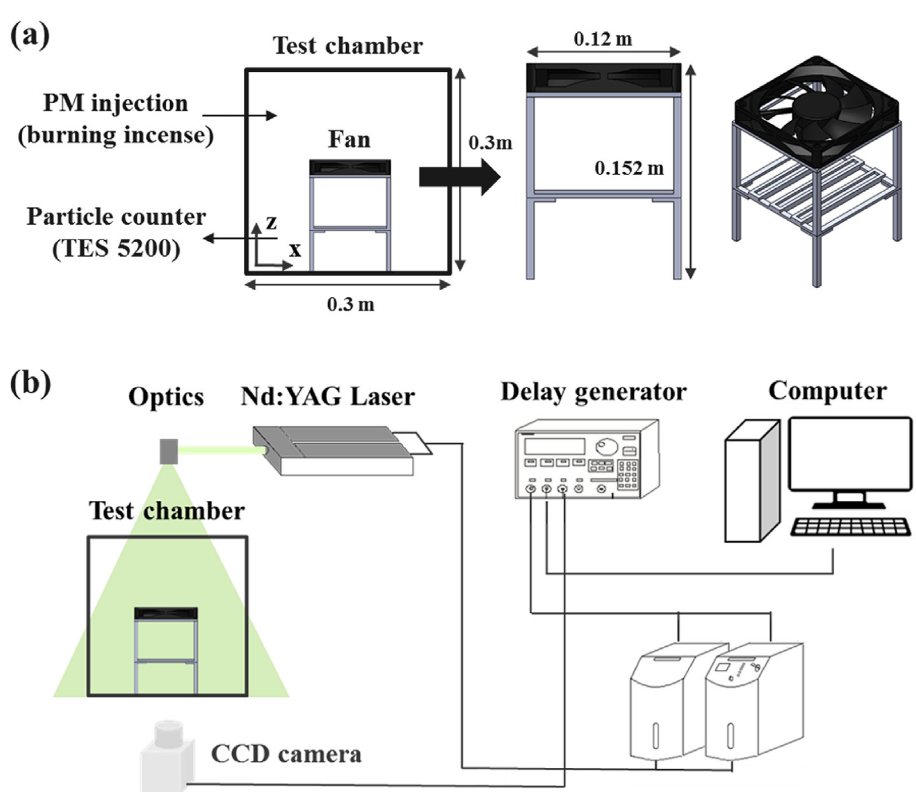

Fig. 1. Schematics of (a) the closed test chamber with flow-inducing experimental setup for PM measurement and (b) the PIV measurement. The PIV system consists of a 120 mJ Nd:YAG laser, a 4K × 3K CCD camera, a delay generator, and the data processor.

algorithm with an interrogation window of 64 × 64-pixel (50% overlapping) based on fast Fourier transform (FFT) was adopted to obtain instantaneous velocity information.

3. Result and discussion

Variation in PM_{10} and $PM_{2.5}$ concentrations inside the chamber were measured for downward flow direction under dry and humid conditions (Fig. 2). PM_{10} concentration was reduced by 30% of initial concentration after 2 h under without flow condition. Only 10% of $PM_{2.5}$ particle concentration reduced after 2 h under without flow condition. PM_{10} particle concentration is mainly affected by gravitational sedimentation (Lai, 2002; Thatcher et al., 2002). For the PM_{10} particles, the concentration decreases rapidly within a short period due to the effect of gravitational sedimentation. Riley et al. (2002) reported that the deposition constant increased when particle diameter ranges from 10^{-1}–10^{1} μm. This supports that the deposition of PM_{10} is faster than that of $PM_{2.5}$. Thus, it would take more time for the deposition of $PM_{2.5}$ under without flow condition.

Under the same RH conditions, the final concentrations of both PM particles decreased with increasing flow velocity (RPM of fan). For direct comparison of PM depositions under dry and humid conditions, the RH difference between dry and humid condition in the test chamber was maintained at higher than 35% for 2 h (Fig. 2c). For both PM_{10} and $PM_{2.5}$ particles, the deposition rate was higher, and the final concentration was lower under humid conditions than under dry conditions at the same flow properties. Differences in the final particle concentrations of $PM_{2.5}$ were more obvious compared with those of PM_{10} particles.

Temporal variations in PM particle concentrations with upward flow direction under dry and humid conditions were observed (Fig. 3). For upward flow, the final concentrations of both PM_{10} and $PM_{2.5}$ were much decreased compared with those for downward flow. Both PMs are largely decreased under humid conditions, compared with those under dry conditions at the same flow conditions. To compare the PM depositions under flow and RH conditions quantitatively, the removal efficiency, η was evaluated using the following formula:

$$\text{Removal efficiency}, \eta = \left(1 - C_f \Big/ C_i\right) * 100(\%) \tag{1}$$

C_i is the initial concentration of PM, and C_f is the final concentration of PM after 2 h. The same data of removal efficiency were used for both upward and downward flow direction at RPM = 0, because the experimental condition was the same regardless of flow direction.

The removal efficiencies of PM particles for downward and upward flows were compared under dry and humid conditions in Fig. 4. Under the same RH conditions and flow directions, the removal efficiency increased with increasing fan RPM for both PM cases. The removal efficiency under humid condition was higher than that under dry condition when other conditions were the same. The maximum difference in $PM_{2.5}$ removal efficiencies between dry and humid conditions was approximately 46% under no flow condition. In addition, the difference in removal efficiencies between humid and dry conditions decreased with increasing RPM. The removal efficiency under upward flow condition was relatively

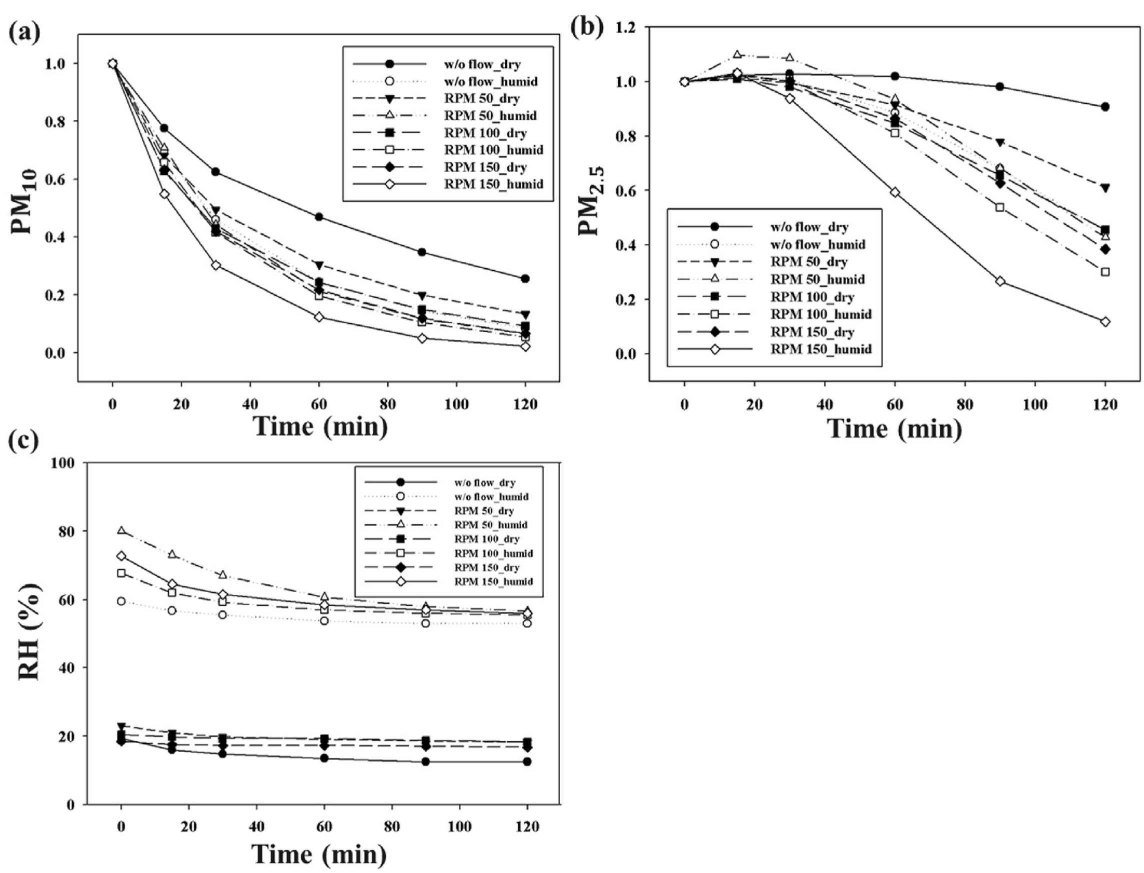

Fig. 2. Temporal variations in (a) PM_{10} and (b) $PM_{2.5}$ particle concentrations of burning incense in the chamber with downward flow under dry and humid conditions. The y-axes represent PM_{10} and $PM_{2.5}$ particle concentrations relative to their initial particle concentrations, respectively. (c) Temporal variations of RH inside the test chamber under dry and humid conditions for 2 h.

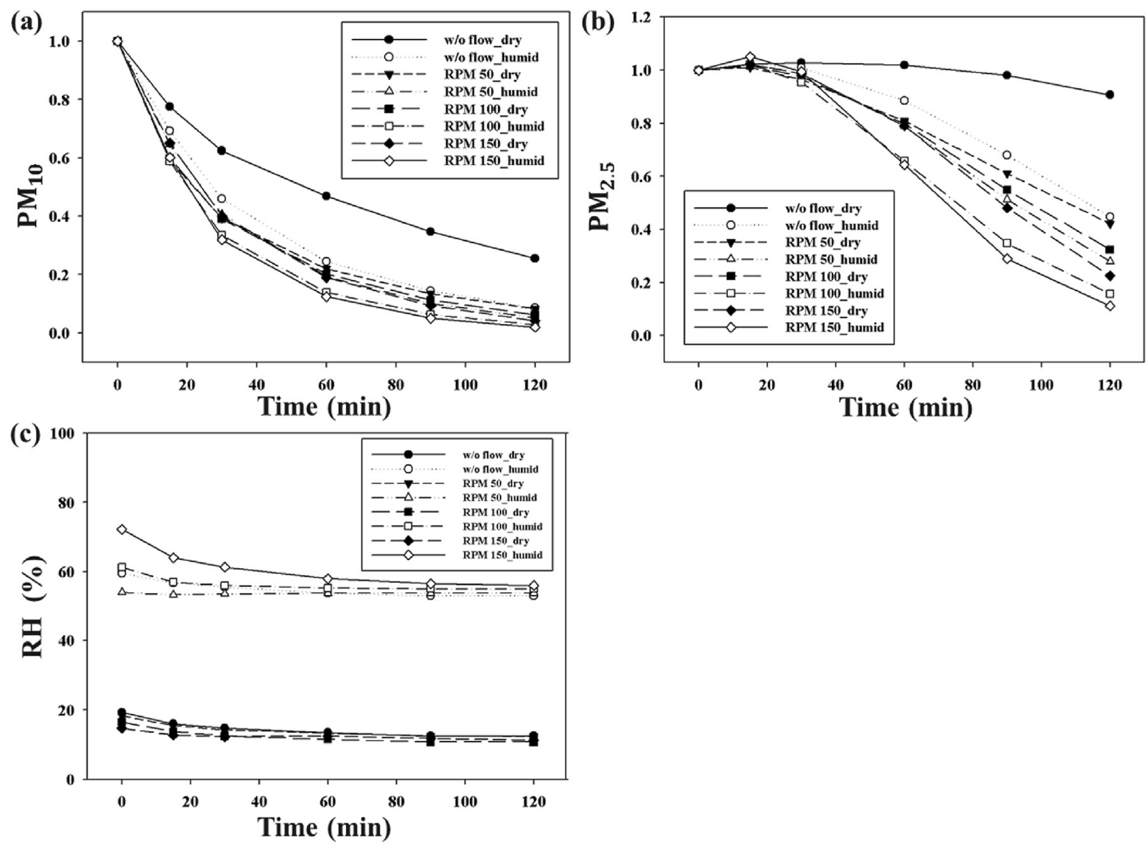

Fig. 3. Temporal variations in (a) PM_{10} and (b) $PM_{2.5}$ particle concentrations of burning incense in the chamber with upward flow under dry and humid conditions. The y-axes represent PM_{10} and $PM_{2.5}$ particle concentrations relative to their initial concentrations, respectively. (c) Temporal variations of RH inside the test chamber under wet and dry conditions for 2 h.

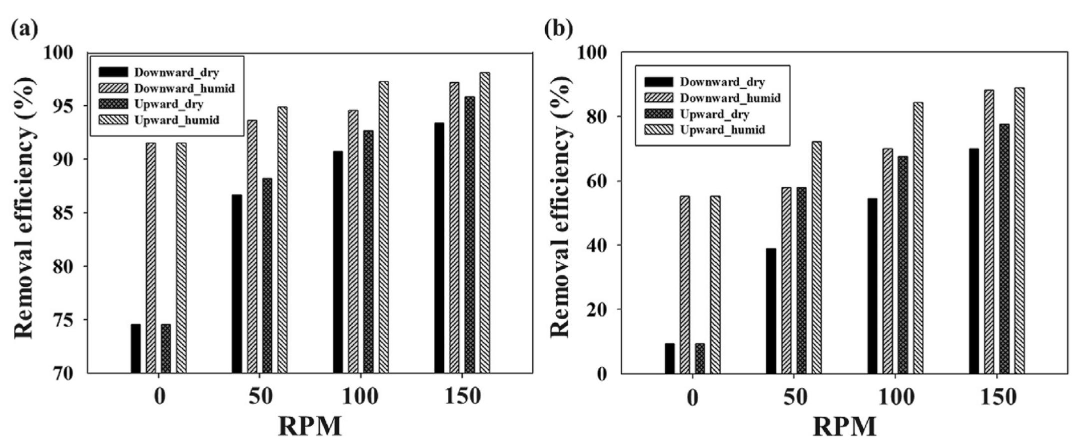

Fig. 4. Comparison of (a) PM_{10} and (b) $PM_{2.5}$ particle removal efficiencies in the chamber for downward flow under dry and humid conditions. The removal efficiency was calculated by comparing the temporal changes in particle concentration from the initial particle concentration in the closed chamber after 2 h.

higher than that under the downward flow condition at the same RH and RPM. The maximum removal efficiencies of $PM_{2.5}$ and PM_{10} under humid and upward flow conditions at RPM = 150 were 88.8% and 98.1%, respectively. The minimum removal efficiencies of $PM_{2.5}$ and PM_{10} under dry condition are observed as 38.8% and 86.7%, respectively, for downward flow direction at RPM = 50.

Considering the lack of air exchange, the deposition constant (λ) was obtained by exponential fitting of PM concentration curve using the following equation:

$$C(t) = C_i(t)\exp(\lambda t) \tag{2}$$

Only values obtained with a correlation coefficient from exponential fitting higher than 95% were retained. The deposition constant (λ) depended on the depositional environment such as surface area, surface roughness, flow condition, surface charge, and surface-to-air temperature difference (Abadie et al., 2001; Lai, 2002). The deposition constant was measured with varying flow and RH conditions inside the test chamber. Deposition constant values of PM_{10} and $PM_{2.5}$ in the chamber for dry and humid conditions under upward and downward flow directions were compared in Fig. 5. The deposition constant increased with increasing fan RPM, and it also increased with upward flow

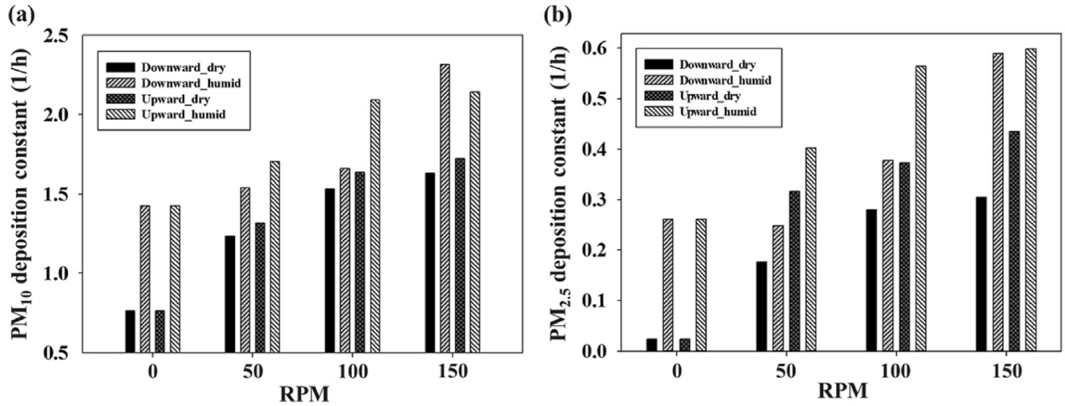

Fig. 5. Comparison of deposition constant of (a) PM_{10} and (b) $PM_{2.5}$ in the chamber for dry and humid conditions under upward and downward flow directions.

direction or under humid conditions. The deposition constant under humid conditions for downward flow at RPM = 150 has the maximum values of 2.32 and 0.644 for PM_{10} and $PM_{2.5}$, respectively. Because the difference in the deposition rate varies depending on the particle diameter, the maximum deposition constant value of PM_{10} are 3.6 times higher than that of $PM_{2.5}$. This result was similar with that obtained in a previous study, in which the deposition constant increased with indoor flow rate (Thatcher et al., 2002).

The deposition constant under humid conditions was significantly different from that under dry conditions for all flow conditions. The deposition characteristics of PM particles are associated with the particle type respond to humidity. This result is intimately associated with the hygroscopic characteristic of particles. The incense PM generated by burning is comprised with hydrophilic oxidized chemical products. Hydrophilic particles are easily agglomerated by water molecules with the large intermolecular attractive force, thus, the size of smoke particles increased due to particle agglomeration (Mikhailov et al., 2006). Li et al. (1992) had examined that RH in the system plays the dominant role in determining the hygroscopic growth of aerosol both experimentally and theoretically. The average growth ratio which defined as the average diameter of wet particles divided by that of dry particles was measured as 1.67 for incense smoke particles (Li and Hopke, 1993). Thus, the hygroscopic growth of particle augments the deposition rate of PM particles (Montgomery et al., 2015).

A PIV experiment was carried out to study the effect of flow condition on PM deposition. Spatial distribution of mean velocity field over the fan positioned at the center of the chamber is shown in Fig. 6 as a function of RPM and flow direction. The flow passes

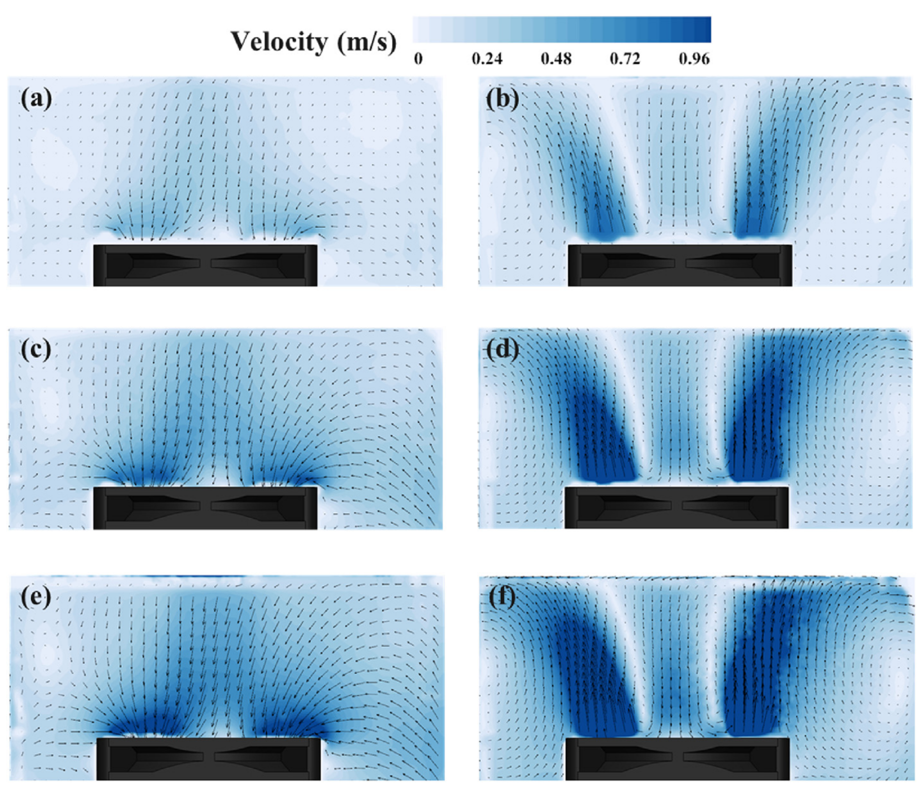

Fig. 6. Comparison of mean velocity fields around the region over the fan inside the test chamber for various RPM and flow direction conditions in the x-z center plane. Downward flow at RPM = (a) 50, (c) 100, (e) 150. Upward flow at RPM = (b) 50, (d) 100, (f) 150.

through the circular ring-shaped region that includes the blade part at top view. For both flow directions, flow speed increased with increasing proximity to the fan. The downward or upward flows were generated by reversing the fan, and the overall mean velocity in the test chamber increased with increasing RPM. The maximum velocity inside the chamber is different between inlet and outlet of fan. The maximum outward and inward flow velocity of fan at RPM = 150 was calculated as 2.3 m/s and 1.2 m/s, respectively.

A comparison of the spatial distributions of the mean velocity field in the region under the fan inside the test chamber is shown in Fig. 7. The overall velocity in the test chamber increased with increasing RPM. This enhanced internal flow increased the removal efficiency and deposition constant regardless of flow direction.

Furthermore, the flow structures inside the chamber are different under upward and downward flow conditions as shown in Figs. 6 and 7. A flow structure was formed in the vertical direction under upward flow condition (Fig. 6b, d, and f). Relatively high-speed flow was formed in the region above the rotating fan blade. The flow moved to the corners of the chamber's upper wall, and then, a counter-rotating vortex pair formed on both sides of the fan under upward flow condition.

A downward diagonal flow was formed under downward flow condition due to the position of the rotating fan and supporting frame structure (Fig. 7a, c, and e). The region under the rotating fan blade had relatively high velocity values. The flow moved to the corners of the bottom wall inside the chamber. A counter-rotating vortex pair opposite to the upward flow condition was formed on both sides of fan under downward flow condition.

The deposition rate is closely related to the vertical flow velocity on the surface (Byrne et al., 1995; Lai, 2002). The mean vertical velocity component (u_v) extracted from the mean velocity field over the fan (Fig. 6) was compared in Fig. 8. The vertical velocity result was extracted along the y-axis with 30 mm distance from the fan. The maximum vertical velocity increases to 0.87, 1.58, and 2.15 m/s as the RPM changes to 50, 100, and 150. For each RPM, the average vertical velocity increased to 0.20, 0.41 and 0.54 m/s under upward flow condition. Under upward flow condition, flow in the vertical direction with a relatively strong momentum was applied to the upper wall of the chamber. This difference increases the number of impaction between PM and the surface of the chamber, and results the increase in PM deposition.

The vertical velocity components with respect to the bottom

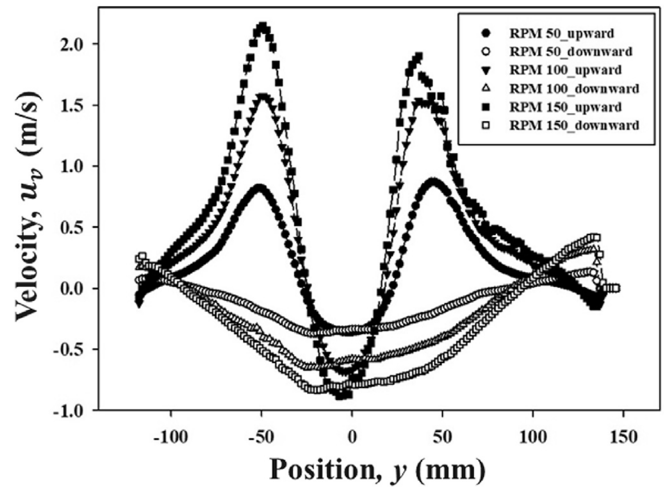

Fig. 8. Comparison of the vertical velocity components (u_v) in the region over the fan inside the test chamber for various RPM and flow direction conditions along the y-axis.

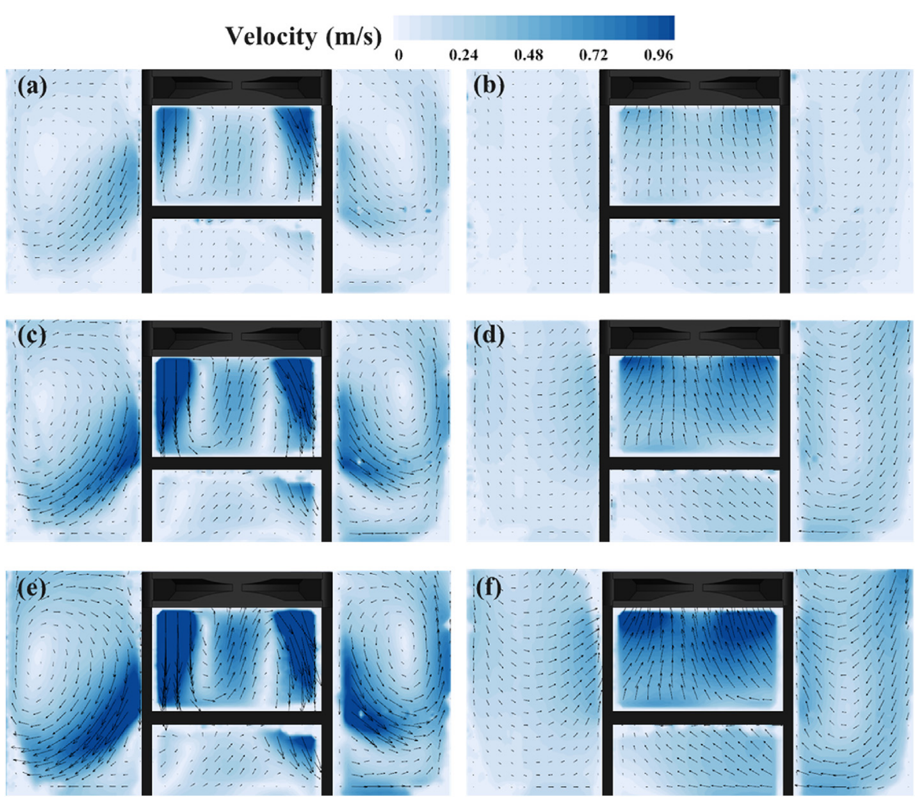

Fig. 7. Comparison of mean velocity fields around the region under the fan inside the test chamber for various RPM and flow direction conditions in the x-z center plane. Downward flow at RPM = (a) 50, (c) 100, (e) 150. Upward flow at RPM = (b) 50, (d) 100, (f) 150.

surface under downward flow condition exhibits smaller than that with respect to the upper surface under upward flow condition due to the location of the fan. Because the distance between the fan and the bottom surface is relative long, the diagonal flow structure is formed under the fan as shown in Fig. 7. Thus, the diagonal flow has relative small value of vertical velocity component to the bottom surface, and it results smaller PM deposition compared with the case under upward flow condition. Therefore, these internal flow structures have influence on the PM removal efficiency and PM deposition constant. A further study on the deposition of PM particles for various fan positions and configurations of the test chamber is required.

4. Conclusion

Indoor PM deposition of incense particles was experimentally investigated with varying internal flow condition including flow rate and flow direction under dry and humid conditions in a closed test chamber. With and without internal flow condition, PM concentrations and removal efficiencies were compared under dry and humid conditions. The removal efficiency and deposition constant of PM increased with increasing fan RPM under the same RHs and flow directions.

The removal efficiency and deposition constant of PM were higher under humid conditions than under dry conditions. This finding supports that the PM deposition effect of RH is associated with hygroscopic property and the size of particles. For upward flow direction at RPM = 150, the maximum removal efficiencies of $PM_{2.5}$ and PM_{10} under humid condition are achieved as 88.8% and 98.1%, respectively. For downward flow direction at RPM = 50, the minimum removal efficiencies of $PM_{2.5}$ and PM_{10} under dry condition are observed as 38.8% and 86.7%, respectively. The maximum deposition constants under humid conditions for downward flow at RPM = 150 were 2.32 and 0.644 for PM_{10} and $PM_{2.5}$, respectively.

Whole velocity fields of the flow inside the chamber were visualized quantitatively using a PIV technique to investigate the effect of internal flow conditions on PM deposition. On one hand, upward flow in the vertical direction with a relatively high momentum was applied to the upper wall of the chamber. On the other hand, the downward flow condition led to diagonal flow's movement toward the corners of the bottom wall.

The present results implied that the internal flow structure and the increase of RH inside the chamber are significant for deposition of PM particles. These results will be utilized as basic information in designing an effective indoor air-purifying device.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.envpol.2019.113263.

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